

Hipotensão, Disfunção Renal e Oligúria no Perioperatório

Professor (a):

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Diarista da UTI da Gastroenterologia do HCFMUSP



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Cirurgias não cardíacas

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Qual a relevância da hipotensão ?

Lesão renal aguda

Injúria miocárdica

Tempo UTI

Sobrevida



Qual a relevância da hipotensão ?

Relationship between Intraoperative Mean Arterial Pressure and Clinical Outcomes after Noncardiac Surgery

Toward an Empirical Definition of Hypotension

Michael Walsh, M.D.,* Philip J. Devereaux, M.D., Ph.D.,† Amit X. Garg, M.D., Ph.D.,‡
Andrea Kurz, M.D.,§ Alparslan Turan, M.D.,|| Reitze N. Rodseth, M.D.,# Jacek Cywinski, M.D.,**
Lehana Thabane, Ph.D.,†† Daniel I. Sessler, M.D.‡‡

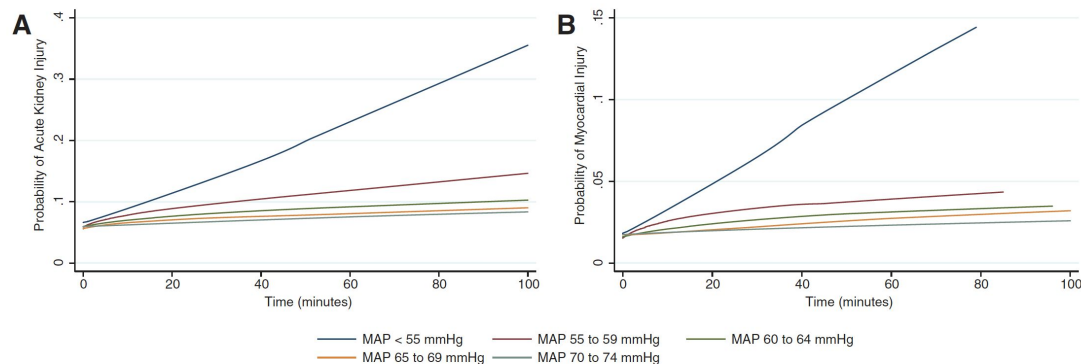


Fig. 2. Predicted risk of (A) acute kidney injury and (B) myocardial injury for each minute the mean arterial pressure (MAP) is <55 mmHg, between 55 and 59 mmHg, between 60 and 64 mmHg, between 65 and 69 mmHg, and between 70 and 74 mmHg during noncardiac surgery. The risk for time in each blood pressure strata is adjusted for time in all other blood pressure strata.

Qual a relevância da hipotensão ?

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Marco de PAM < 55 mmHg foi muito relevante para os desfechos negativos.



Qual a relevância da hipotensão ?

Association of Intraoperative Hypotension with Acute Kidney Injury after Elective Noncardiac Surgery

Louise Y. Sun, M.D., S.M., Duminda N. Wijeyesundera, M.D., Ph.D., Gordon A. Tait, Ph.D., W. Scott Beattie, M.D., Ph.D.

Table 2. Proportion Experiencing AKI, Stratified by Hypotension Duration for MAP Thresholds of 55, 60, and 65 mmHg

IOH Duration (min)	MAP < 55 mmHg		MAP < 60 mmHg		MAP < 65 mmHg	
	N	AKI	N	AKI	N	AKI
1–5	2,807	189 (6.7%)	2,490	137 (5.5%)	1,474	64 (4.3%)
6–10	637	63 (9.9%)	1,030	64 (6.2%)	1,252	79 (6.3%)
11–20	63	7 (11.1%)	579	67 (11.6%)	1,182	80 (6.8%)
>20	23	4 (17.4%)	274	30 (11.0%)	903	92 (10.2%)

AKI = acute kidney injury; IOH = intraoperative hypotension; MAP = mean arterial pressure.



Qual a relevância da hipotensão ?

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PAM < 65 mmHg em qualquer momento;
queda de 20% basal; PAM < 50 mmHg
mesmo por 1 minuto.

=

Piores desfechos



HIPO TENSÃO INTRAOPERATÓRIA X DISFUNÇÃO ORGÂNICA

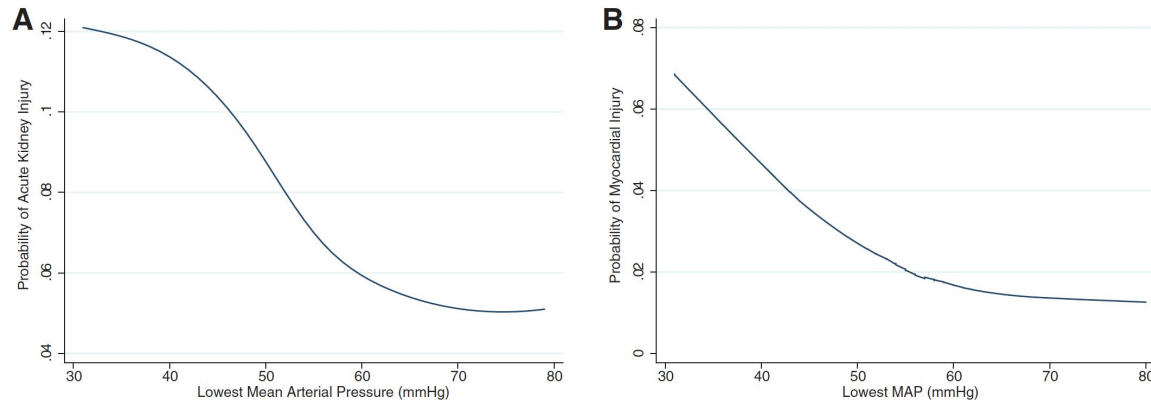


Fig. 3. Predicted probability of (A) acute kidney injury and (B) myocardial injury by lowest mean arterial pressure (MAP) experienced during surgery.

Toward an Empirical Definition of Hypotension. *Anesthesiology* 2013; 119:507–515 doi:
<https://doi.org/10.1097/ALN.0b013e3182a10e26>



HIPO TENSÃO x RIM

Heterogeneity and Variability in Pressure Autoregulation of Organ Blood Flow: Lessons Learned Over 100+ Years

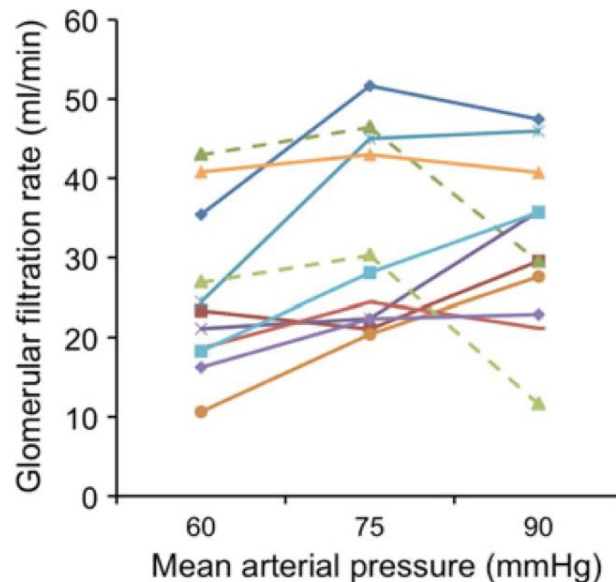
Lingzhong Meng, MD¹; Yingwei Wang, MD, PhD²; Lina Zhang, MD³; David L. McDonagh, MD⁴

Table 4 Effects of norepinephrine on renal variables

	60 mmHg	→	75 mmHg
RBF (ml/min)	442 ± 156		477 ± 178
RDO ₂ (ml/min)	59.9 ± 19.0	→	67.9 ± 23.8*
RVR (mmHg/ml/min)	0.126 ± 0.046		0.146 ± 0.051*
GFR (ml/min)	25.3 ± 10.3	→	32.3 ± 11.8**

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ 75 mmHg versus 60 mmHg, ### $p < 0.001$ 90 mmHg versus 75 mmHg

RDO2 (DO2 renal)



Sugestão

- Use PAM ≥ 60 mmHg como limiar, evitando >10 min abaixo de 55 mmHg.
- Trate IOH rapidamente conforme o mecanismo (pré-carga/contratilidade/vasoplegia).
- Atenção a sangramento e escolha de fluidos (preferir soluções balanceadas).



HIPO TENSÃO INTRA OPERATÓRIA AUMENTA
MORTALIDADE ?



Intraoperative Mean Arterial Pressure Variability and 30-day Mortality in Patients Having Noncardiac Surgery

Edward J. Mascha, Ph.D., Dongsheng Yang, M.S., Stephanie Weiss, M.D., Daniel I. Sessler, M.D.

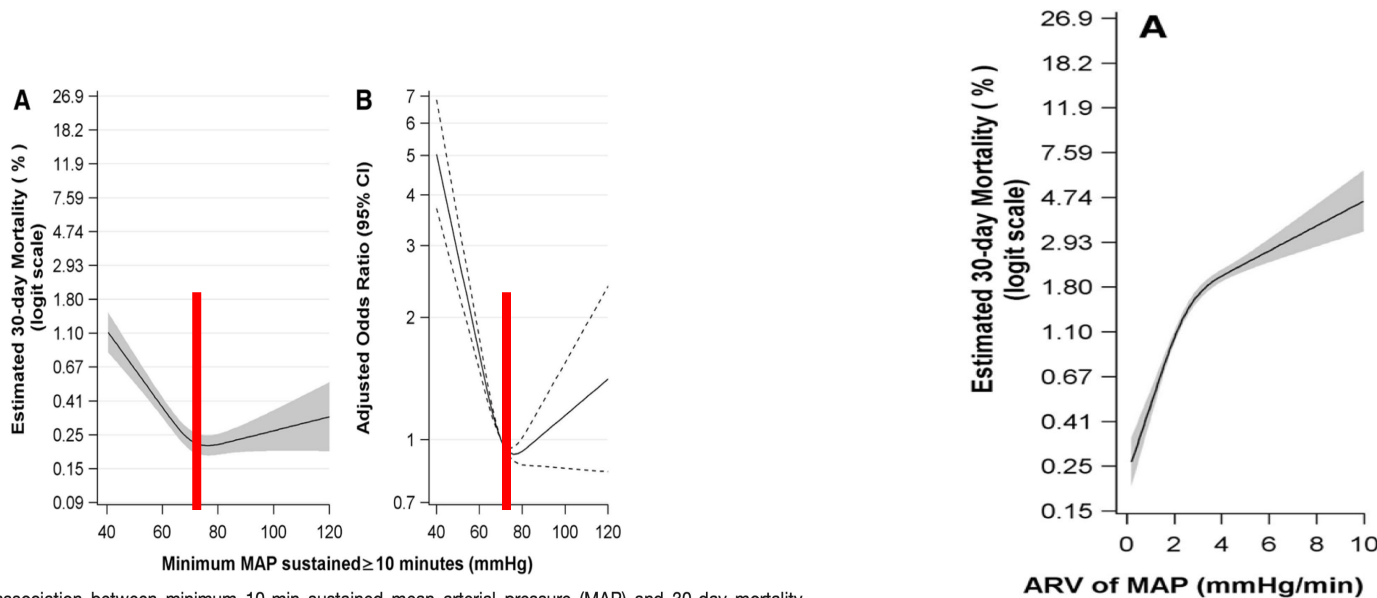


Fig. 5. Multivariable association between minimum 10-min sustained mean arterial pressure (MAP) and 30-day mortality.

PAM média IOP x Mortalidade

Variabilidade de PAM x Mortalidade



IRA perioperatória



INTRODUÇÃO

Postoperative renal dysfunction after noncardiac surgery

Suvi T. Vaara^a and Rinaldo Bellomo^{a,b}

“Redução aguda da TFG na 1ª semana após a cirurgia”



Importância

- Em pacientes de alto risco >> IRA 15 - 40%
- Maior tempo de internação e morbimortalidade
- 300.000 mortes/ano associadas à IRA perioperatória
- Incidência de 300.000 casos de DRC a cada ano associadas à IRA perioperatória

J Am Soc Nephrol 2011; 22:14–20
Kidney International (2017) 92, 721–728



Importância

Variations in the Risk of Acute Kidney Injury Across Intraabdominal Surgery Procedures

Minjae Kim, MD, MS,* Joanne E. Brady, PhD,*† and Guohua Li, MD, DrPH*†



pixtastock.com - 48842426

https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.pixtastock.com%2Fillustration%2F4884242&psig=AOvVaw0qeD_-xu_Kbz3hhfQOSb2f&ust=1677334764586000&source=images&cd=vfe&ved=0CA0QjRxqFwoTCIC464utr0CFQAAAAAABAI

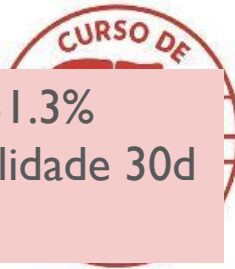
PO chirurgia Abd

Sem LRA

1.9 %
Mortalidade 30d

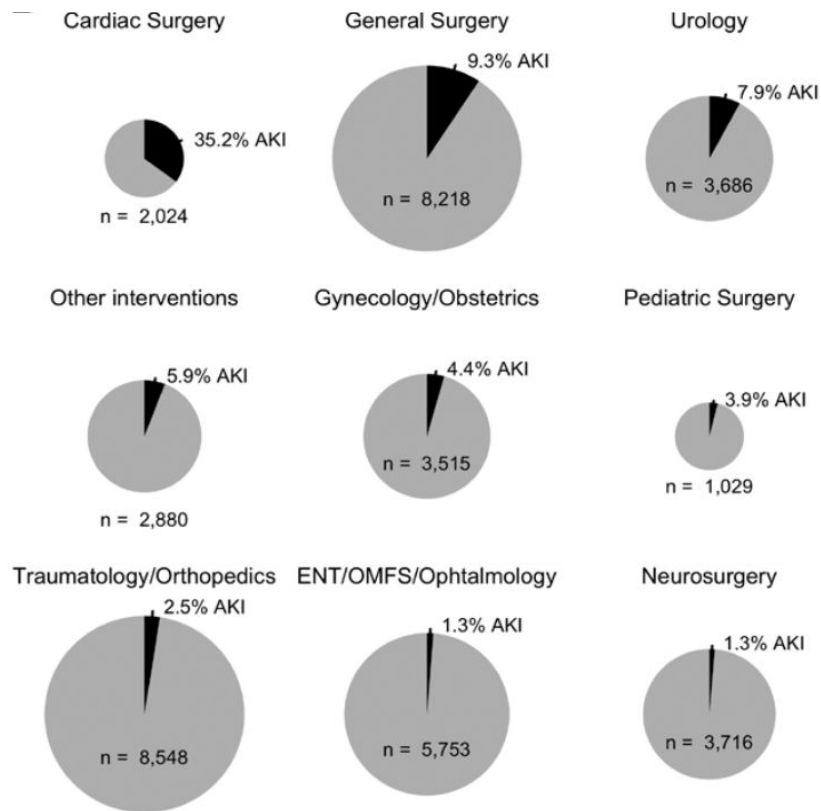
LRA

31.3%
Mortalidade 30d



Incidência de acordo com especialidade cirúrgica:

- Cardíaca - 35,2%
- Geral - 9,3%
- Urologia - 7,9%
- Gineco/obs 4.4%
- Ortopedia 2,5%
- Neurocirurgia 1.3%



Etiologia multifactorial

Preoperative factors

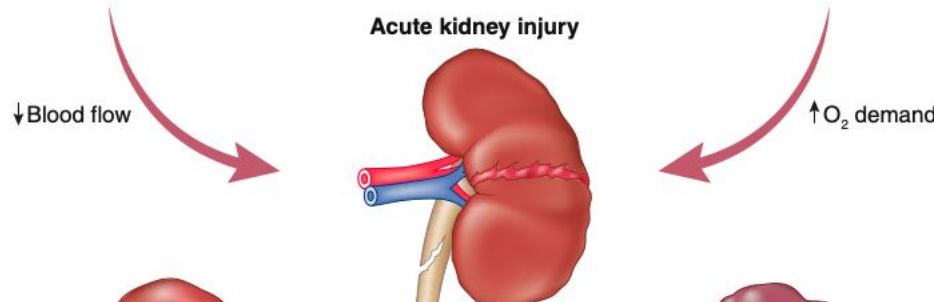
- Preexisting kidney dysfunction
- Diabetes
- Cardiac dysfunction
- Age
- Sepsis
- Volume depletion
- Hepatic failure
- Crush injury
- Exposure to nephrotoxins

Intraoperative factors

- Hypovolemia (caused by bleeding and insensible fluid losses)
- Kidney ischemia
- Inflammation
- Increased intra-abdominal pressure
- Decreased cardiac output (caused by anesthetic)
- Vasodilatation (caused by anesthetic)
- Exposure to nephrotoxins
- Embolism

Postoperative factors

- Hypovolemia (caused by bleeding and insensible fluid losses)
- Kidney ischemia
- Inflammation
- Increased intra-abdominal pressure
- Decreased cardiac output (caused by anesthetic)
- Vasodilatation (caused by anesthetic)
- Exposure to nephrotoxins
- Urinary obstruction
- Acute lung injury
- Mechanical ventilation



Diagnóstico

- Escores baseados em diurese e creatinina;

Falta de consenso

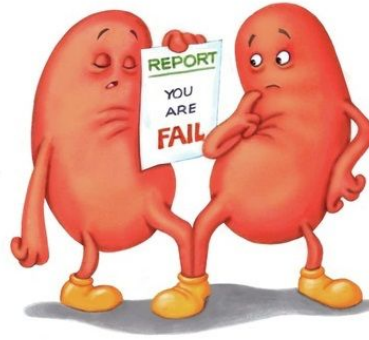
(KDIGO ?)

Estágio	Medições para lesão renal aguda (qualquer um dos seguintes para cada estágio)	
	Aumento da creatinina sérica	Declínio na quantidade do débito urinário
1	$\geq 0,3$ mg/dL (26,52 micromol/L) ou 1,5–1,9 vezes a linha de base	$< 0,5$ mL/kg/h por 6–12 horas
2	2 a 2,9 vezes a linha de base	$< 0,5$ mL/kg/hora por ≥ 12 horas
3	$\geq 4,0$ mg/dL (353,60 micromol/L) ou ≥ 3 vezes a linha de base	$< 0,3$ mL/kg/h por ≥ 24 horas ou anúria por ≥ 12 h



Redução de Débito Urinário / Oligúria

- Adaptativa ?
 - REMIT / SIRS
- Patológica ?
 - Hipovolemia / sangramento / hipotensão / PIA



Oligúria

Importance of intraoperative oliguria during major abdominal surgery: findings of the Restrictive Versus Liberal Fluid Therapy in Major Abdominal Surgery trial

Paul S. Myles^{1,2,*}, David R. McIlroy^{1,2}, Rinaldo Bellomo^{3,4} and Sophie Wallace^{1,2}

36,7% fizeram oligúria no IOP

75% não resultou em IRA laboratorial

VPP da oligúria apenas 25%



Oligúria

Importance of intraoperative oliguria during major abdominal surgery: findings of the Restrictive Versus Liberal Fluid Therapy in Major Abdominal Surgery trial

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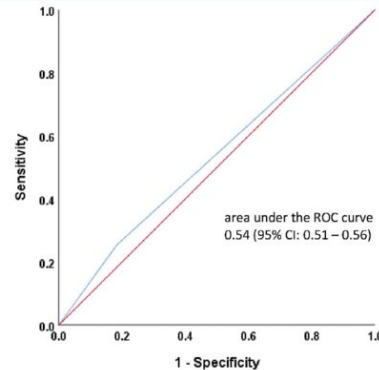


Fig 3. Receiver operating characteristic (ROC) curve depicting the poor predictive utility of intraoperative oliguria. CI, confidence interval.

FAIL



Oligúria

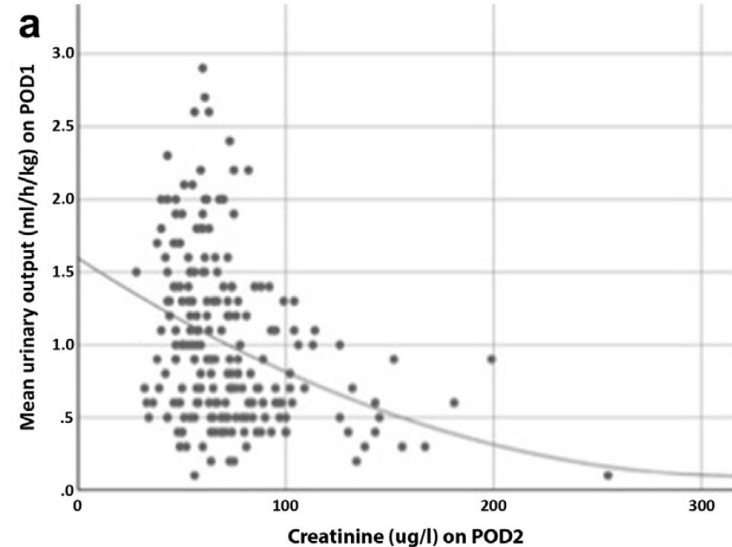
ORIGINAL ARTICLE

Acute kidney injury after liver surgery: does postoperative urine output correlate with postoperative serum creatinine?

Gaëtan-Romain Joliat, Ismail Labgaa, Nicolas Demartines & Nermin Halkic

Department of Visceral Surgery, Lausanne University Hospital CHUV, Lausanne, Switzerland

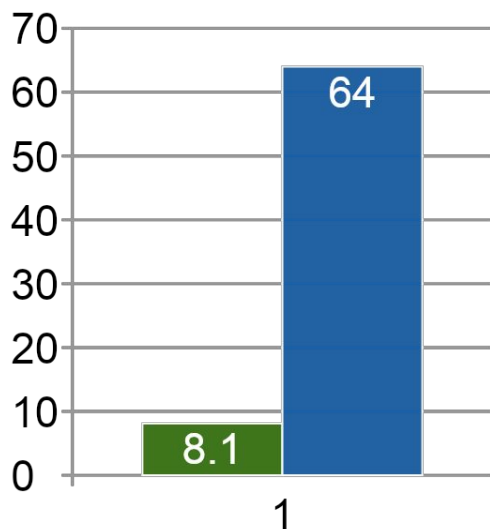
**Correlação fraca entre
Diurese 1 PO e Creatinina 2 PO**



IRA oligúrica X IRA não oligúrica

Coorte 4229 pacientes submetidos à cirurgia de grande porte

Incidência (%)

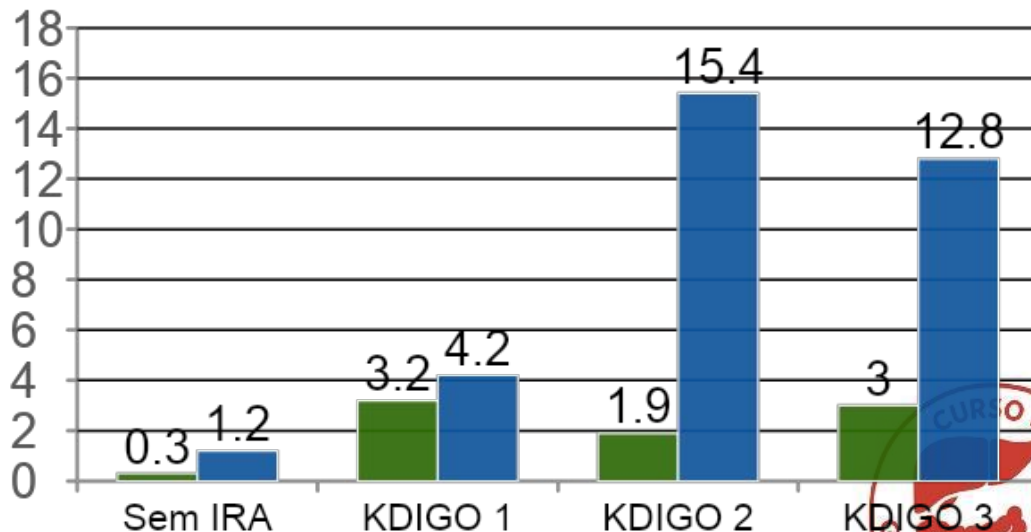


■ Creatinina

■ Diurese + Creatinina

Critério

Mortalidade (%)



É possível prevenir ?



Interventions for protecting renal function in the perioperative period (Review)

Zacharias M, Mugawar M, Herbison GP, Walker RJ, Hovhannisyan K, Sivalingam P, Conlon NP

DATA AND ANALYSES	
Analysis 1.1. Comparison 1 Dopamine and analogues versus no intervention, Outcome 1 Mortality.	parison 7 Erythropoietin (EPO) versus control, Outcome 4 Urine output: 2 to 3 days.
Analysis 1.2. Comparison 1 Dopamine and analogues versus no intervention, Outcome 2 Acute renal injury.	parison 7 Erythropoietin (EPO) versus control, Outcome 5 Urine output: 5 to 7 days.
Analysis 1.3. Comparison 1 Dopamine and analogues versus no intervention, Outcome 3 Urine output.	parison 8 Intravenous fluid versus control, Outcome 1 Mortality.
Analysis 1.4. Comparison 1 Dopamine and analogues versus no intervention, Outcome 4 Creatinine clearance.	parison 8 Intravenous fluid versus control, Outcome 2 Acute renal injury.
Analysis 1.5. Comparison 1 Dopamine and analogues versus no intervention, Outcome 5 Free water clearance.	parison 8 Intravenous fluid versus control, Outcome 3 Creatinine clearance.
Analysis 1.6. Comparison 1 Dopamine and analogues versus no intervention, Outcome 6 Fractional excretion of sodium.	parison 9 Cardiac surgery: subgroup analysis, Outcome 1 Mortality.
Analysis 1.7. Comparison 1 Dopamine and analogues versus no intervention, Outcome 7 Renal plasma flow (24 hours).	parison 9 Cardiac surgery: subgroup analysis, Outcome 2 Acute renal injury.
Analysis 2.1. Comparison 2 Diuretics versus no intervention, Outcome 1 Mortality.	parison 9 Cardiac surgery: subgroup analysis, Outcome 3 Urine output.
Analysis 2.2. Comparison 2 Diuretics versus no intervention, Outcome 2 Acute renal injury.	parison 9 Cardiac surgery: subgroup analysis, Outcome 4 Creatinine clearance.
Analysis 2.3. Comparison 2 Diuretics versus no intervention, Outcome 3 Urine output.	parison 9 Cardiac surgery: subgroup analysis, Outcome 5 Free water clearance.
Analysis 2.4. Comparison 2 Diuretics versus no intervention, Outcome 4 Creatinine clearance.	parison 9 Cardiac surgery: subgroup analysis, Outcome 6 Fractional excretion of sodium.
Analysis 3.1. Comparison 3 Calcium channel blockers versus no intervention, Outcome 1 Mortality.	parison 10 Aortic surgery: subgroup analysis, Outcome 1 Mortality.
Analysis 3.2. Comparison 3 Calcium channel blockers versus no intervention, Outcome 2 Acute renal injury.	parison 10 Aortic surgery: subgroup analysis, Outcome 2 Acute renal injury.
Analysis 3.3. Comparison 3 Calcium channel blockers versus no intervention, Outcome 3 Urine output.	parison 10 Aortic surgery: subgroup analysis, Outcome 3 Urine output.
Analysis 3.4. Comparison 3 Calcium channel blockers versus no intervention, Outcome 4 Creatinine clearance.	parison 10 Aortic surgery: subgroup analysis, Outcome 4 Creatinine clearance.
Analysis 3.5. Comparison 3 Calcium channel blockers versus no intervention, Outcome 5 Free water clearance.	parison 10 Aortic surgery: subgroup analysis, Outcome 5 Free water clearance.
Analysis 4.1. Comparison 4 ACE inhibitors versus no intervention, Outcome 1 Mortality.	parison 10 Aortic surgery: subgroup analysis, Outcome 6 Fractional excretion of sodium.
Analysis 4.2. Comparison 4 ACE inhibitors versus no intervention, Outcome 2 Acute renal injury.	parison 10 Aortic surgery: subgroup analysis, Outcome 7 Renal plasma flow.
Analysis 4.3. Comparison 4 ACE inhibitors versus no intervention, Outcome 3 Renal plasma flow.	parison 11 Biliary surgery: subgroup analysis, Outcome 1 Urine output.
Analysis 5.1. Comparison 5 Atrial natriuretic peptide versus no intervention, Outcome 1 Mortality.	parison 11 Biliary surgery: subgroup analysis, Outcome 2 Creatinine clearance.
Analysis 5.2. Comparison 5 Atrial natriuretic peptide versus no intervention, Outcome 2 Acute renal injury.	parison 12 Studies on participants with pre-existing renal impairment, Outcome 1 Mortality.
Analysis 5.3. Comparison 5 Atrial natriuretic peptide versus no intervention, Outcome 3 Urine output, 24 hours.	parison 12 Studies on participants with pre-existing renal impairment, Outcome 2 Acute renal injury.
Analysis 5.4. Comparison 5 Atrial natriuretic peptide versus no intervention, Outcome 4 Creatinine clearance, 24 hours.	parison 12 Studies on participants with pre-existing renal impairment, Outcome 3 Urine output.
Analysis 5.5. Comparison 5 Atrial natriuretic peptide versus no intervention, Outcome 5 Creatinine clearance, 2 to 3 days.	parison 12 Studies on participants with pre-existing renal impairment, Outcome 4 Creatinine clearance.
Analysis 6.1. Comparison 6 N-Acetyl cysteine versus no intervention, Outcome 1 Mortality.	parison 13 Studies with low risk of bias: sensitivity analysis, Outcome 1 Reported mortality, low risk of bias.
Analysis 6.2. Comparison 6 N-Acetyl cysteine versus no intervention, Outcome 2 Acute renal injury.	
Analysis 6.3. Comparison 6 N-Acetyl cysteine versus no intervention, Outcome 3 Urine output, 24 hours.	
Analysis 7.1. Comparison 7 Erythropoietin (EPO) versus control, Outcome 1 Mortality.	
Analysis 7.2. Comparison 7 Erythropoietin (EPO) versus control, Outcome 2 Acute renal injury.	
Analysis 7.3. Comparison 7 Erythropoietin (EPO) versus control, Outcome 3 Urine output: 24 hours.	

Interventions for protecting renal function in the perioperative period (Review)

Zacharias M, Mugawar M, Herbison GP, Walker RJ, Hovhannisyan K, Sivalingam P, Conlon NP

AUTHORS' CONCLUSIONS**Implications for practice**

No convincing evidence suggests that pharmacological or other interventions used to protect the kidneys during surgery are of benefit to patients, as seen from 72 included studies. Reasonable

**Perioperative Acute Kidney Injury:
Prevention, Early Recognition, and
Supportive Measures**Stefano Romagnoli^a Zaccaria Ricci^b Claudio Ronco^{c, d}**Renal Replacement Therapy**

In severe AKI, especially in case of fluid accumulation in hemodynamically unstable anuric patients, RRT is the only supportive measure. The best RRT mode, in-



Effect of a Perioperative, Cardiac Output-Guided Hemodynamic Therapy Algorithm on Outcomes Following Major Gastrointestinal Surgery

A Randomized Clinical Trial and Systematic Review

Rupert M. Pearse, MD; David A. Harrison, PhD; Neil MacDonald, FRCA; Michael A. Gillies, FRCA; Mark Blunt, FRCA; Gareth Ackland, PhD; Michael P. W. Grocott, MD; Aoife Ahern, BSc; Kathryn Griggs, MSc; Rachael Scott, PhD; Charles Hinds, FRCA; Kathryn Rowan, PhD; for the OPTIMISE Study Group

Volume para otimizar SV + Dose fixa **inotrópico**

bolus de 250 colóide

Dopexamina

Table 3. Results for the Primary Outcome^a

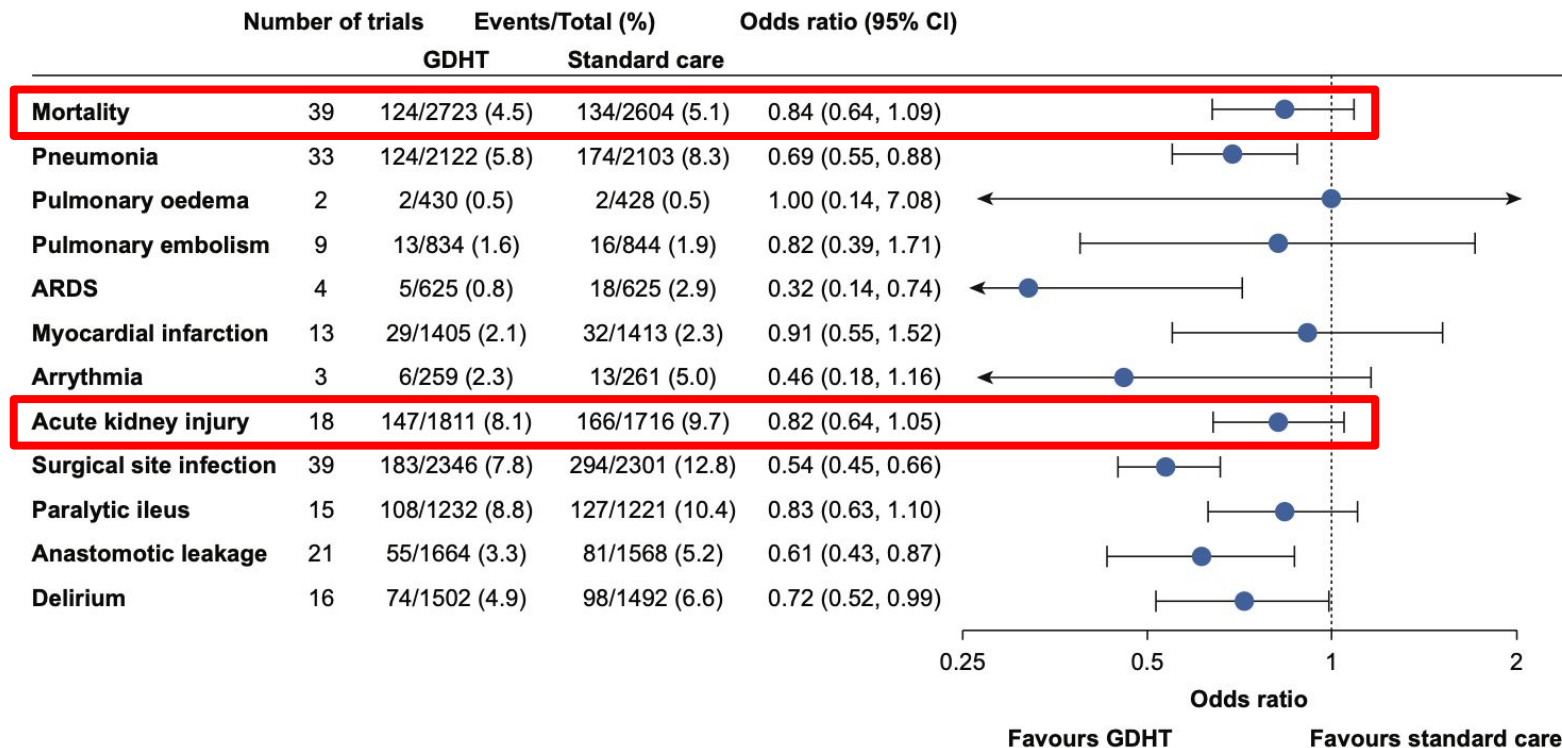
Outcomes	Cardiac Output-Guided Hemodynamic Therapy Algorithm, No. (%) (n = 366)	Usual Care, No. (%) (n = 364)
Composite of predefined moderate or major postoperative complications and mortality at 30 d following surgery ^b	134 (36.6)	158 (43.4)
Individual elements		
Mortality	12 (3.3)	11 (3.0)
Pulmonary embolism	4 (1.1)	1 (0.3)
Myocardial ischemia or infarction	10 (2.7)	8 (2.2)
Arrhythmia	39 (10.7)	40 (11.0)
Cardiac or respiratory arrest	16 (4.4)	14 (3.8)
Limb or digital ischemia	2 (0.5)	1 (0.3)
Cardiogenic pulmonary edema	1 (0.3)	2 (0.5)
Acute respiratory distress syndrome	3 (0.8)	4 (1.1)
Gastrointestinal bleeding	13 (3.6)	8 (2.2)
Bowel infarction	2 (0.5)	5 (1.4)
Anastomotic breakdown	12 (3.3)	16 (4.4)
Paralytic ileus	20 (5.5)	27 (7.4)
Acute psychosis	3 (0.8)	8 (2.2)
Stroke	1 (0.3)	0
Acute kidney injury	17 (4.6)	17 (4.7)
Infection, source uncertain	11 (3.0)	9 (2.5)
Urinary tract infection	9 (2.5)	9 (2.5)
Surgical site infection ^c	22 (6.0)	39 (10.7)
Organ/space infection	20 (5.5)	36 (9.9)
Bloodstream infection	6 (1.6)	15 (4.1)
Nosocomial pneumonia	36 (9.8)	39 (10.7)
Postoperative hemorrhage	6 (1.6)	4 (1.1)
Self-assessment of blinding for outcome assessment ^d		
Assessor suitably blinded	342 (94.2)	349 (96.7)
Assessor may have known allocation	9 (2.5)	6 (1.7)
Assessor knew allocation ^e	12 (3.3)	6 (1.7)

^a Reports complications; some patients developed more than 1 complication. Data do not include 1 patient in the usual care group who was randomized in error and 3 patients (1 in the usual care group and 2 in the hemodynamic therapy group) who withdrew consent. The predefined complication of other infections of the urinary tract did not occur in any patient.

^b Relative risk, 0.84; 95% CI, 0.71-1.01; *P* = .07.

Goal-directed haemodynamic therapy during general anaesthesia for noncardiac surgery: a systematic review and meta-analysis

76 trials



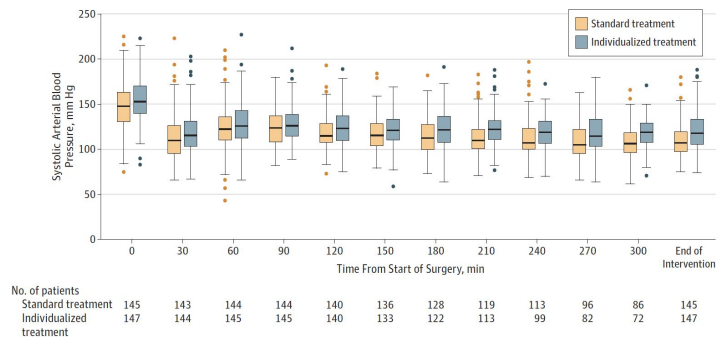
É POSSÍVEL PREVENIR ?

JAMA | Original Investigation | CARING FOR THE CRITICALLY ILL PATIENT

Effect of Individualized vs Standard Blood Pressure Management Strategies on Postoperative Organ Dysfunction Among High-Risk Patients Undergoing Major Surgery A Randomized Clinical Trial

Emmanuel Futier, MD, PhD; Jean-Yves Lefrant, MD, PhD; Pierre-Gregoire Guinot, MD, PhD; Thomas Godet, MD, PhD; Emmanuel Lorne, MD; Philippe Cuvillon, MD, PhD; Sebastien Bertran, MD; Marc Leone, MD, PhD; Bruno Pastene, MD; Vincent Piriou, MD, PhD; Serge Molliex, MD, PhD; Jacques Albanese, MD, PhD; Jean-Michel Julia, MD; Benoit Tavernier, MD, PhD; Etienne Imhoff, MD; Jean-Etienne Bazin, MD, PhD; Jean-Michel Constantin, MD, PhD; Bruno Pereira, PhD; Samir Jaber, MD, PhD; for the INPRESS Study Group

Figure 2. Systolic Arterial Blood Pressure in the Individualized and Standard Treatment Groups Over the Intervention Period



Systolic arterial blood pressures were higher in the individualized treatment group ($P < .001$ by random-effect model for the between-group comparison).

induction to 4 hours after completion of surgery. The median (IQR) duration of surgery was 260 (170-365) minutes in the individualized treatment group and

Efedrina se PAS < 80 ou queda
40% basal (standard)

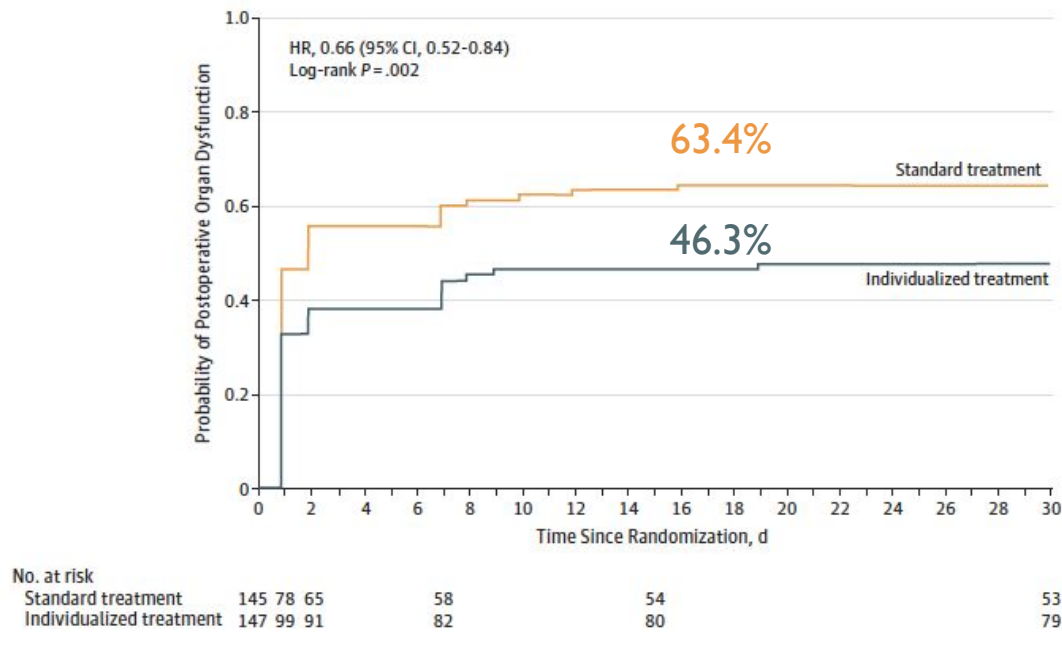
VS.

Nora para manter PAS até 10%
basal



É POSSÍVEL PREVENIR ?

Figure 3. Kaplan-Meier Estimates of the Probability of Postoperative Organ Dysfunction by Day 30 After Surgery



+ Volume ?

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812

JUNE 14, 2018

VOL. 378 NO. 24

Restrictive versus Liberal Fluid Therapy for Major Abdominal Surgery

P.S. Myles, R. Bellomo, T. Corcoran, A. Forbes, P. Peyton, D. Story, C. Christophi, K. Leslie, S. McGuinness, R. Parke, J. Serpell, M.T.V. Chan, T. Painter, S. McCluskey, G. Minto, and S. Wallace, for the Australian and New Zealand College of Anaesthetists Clinical Trials Network and the Australian and New Zealand Intensive Care Society Clinical Trials Group*

Table 2. Blood Loss and Administered Intravenous-Fluid Volumes.*

Variable	Restrictive Fluid (N = 1490)	Liberal Fluid (N = 1493)	P Value
Median intraoperative fluid administration (IQR) — ml			
Crystalloid	1677 (1173 to 2294)	3000 (2100 to 3850)	<0.001
Colloid‡	500 (250 to 800)	500 (400 to 1000)	0.01
Postoperative day 1, post-PACU			
Median administration of fluid (IQR) — ml			
Crystalloid	1556 (1200 to 1960)	2600 (2052 to 3150)	<0.001
Colloid‡	500 (250 to 1000)	500 (400 to 750)	0.89



+ Volume ?

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Table 2. Blood Loss and Administered Intravenous-Fluid Volumes.*

Variable	Restrictive Fluid (N = 1490)	Liberal Fluid (N = 1493)	P Value
At 24 hr after surgery			
Median cumulative total for intravenous fluids (IQR) — ml	3671 (2885 to 4880)	6146 (5000 to 7410)	<0.001



Quanto de volume ?

Table 3. Primary and Secondary Outcomes.*

Outcome	Restrictive Fluid (N=1490)	Liberal Fluid (N=1493)	Hazard or Risk Ratio (95% CI) [†]	P Value
Secundários				
Surgical-site infection — no./total no. (%)	245/1481 (16.5)	202/1487 (13.6)	1.22 (1.03–1.45)	0.02
Acute kidney injury — no./total no. (%)**	124/1443 (8.6)	72/1439 (5.0)	1.71 (1.29–2.27)	<0.001
Renal-replacement therapy — no./total no. (%)	13/1460 (0.9)	4/1462 (0.3)	3.27 (1.01–13.8)	0.048

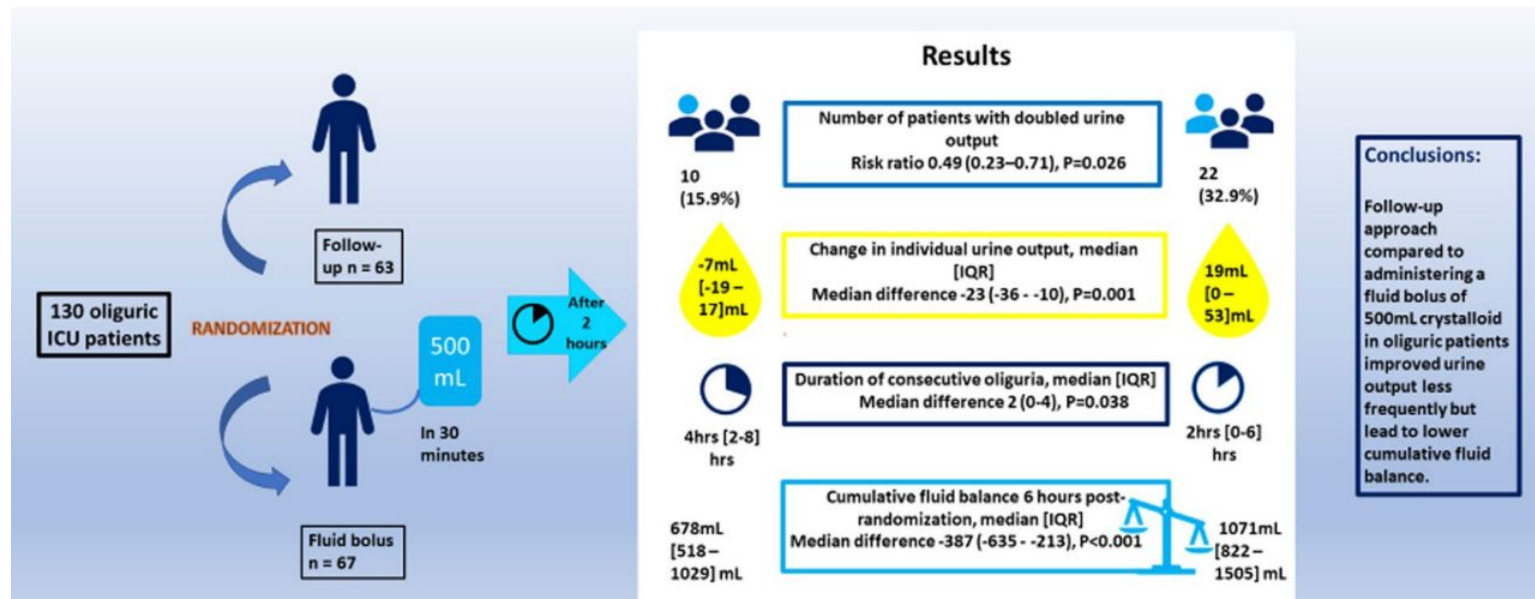




Non-interventional follow-up versus fluid bolus in RESPONSE to oliguria in hemodynamically stable critically ill patients: a randomized controlled pilot trial

Nina Inkinen^{1,2*}, Ville Pettilä², Miia Valkonen², Majja Serlo², Minna Bäcklund², Johanna Hästbacka², Anni Pulkkinen¹, Tuomas Selander³ and Suvi T. Vaara²

TRATAR OLIGÚRIA COM VOLUME?



RESEARCH**Open Access**

The effect of excess fluid balance on the mortality rate of surgical patients: a multicenter prospective study

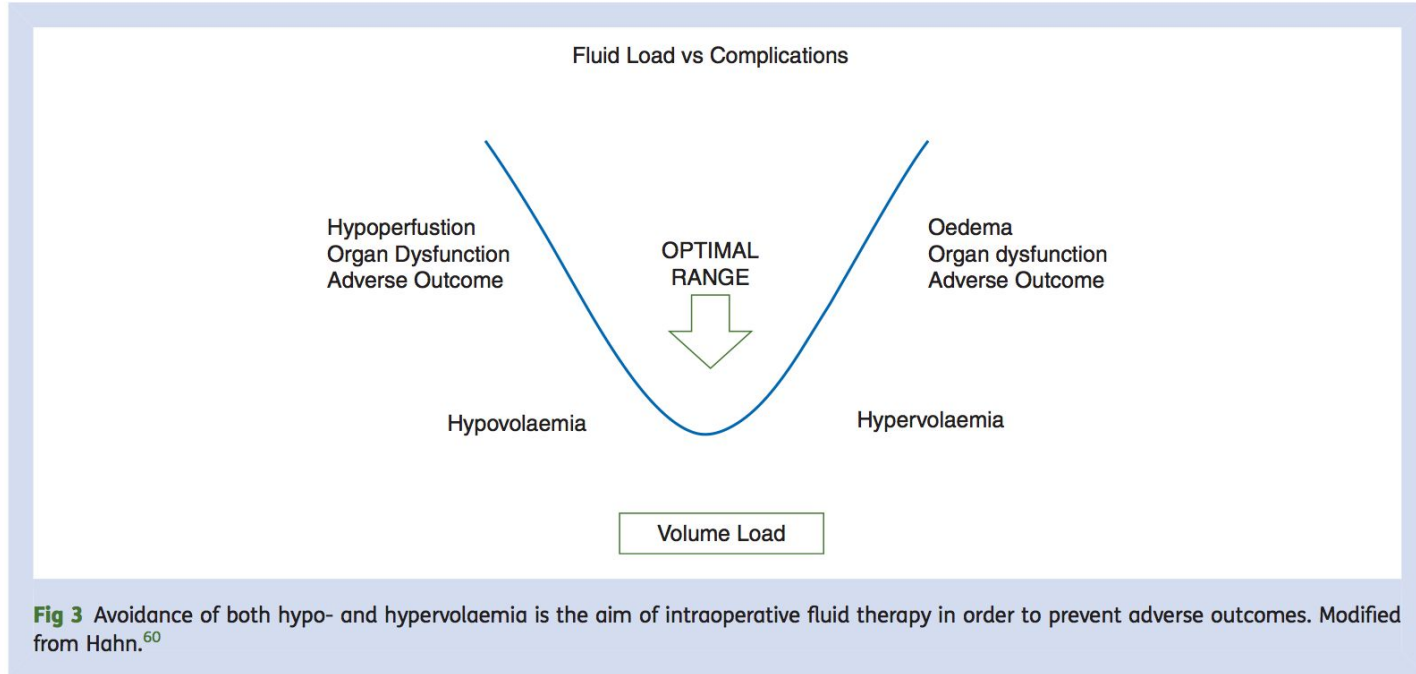
João M Silva Jr^{1,2,4*}, Amanda Maria Ribas Rosa de Oliveira^{2,3}, Fernando Augusto Mendes Nogueira¹, Pedro Monferrari Monteiro Vianna¹, Marcos Cruz Pereira Filho¹, Leandro Ferreira Dias¹, Vivian Paz Leão Maia¹, Cesar de Souza Neucamp¹, Cristina Prata Amendola³, Maria José Carvalho Carmona² and Luiz M Sá Malbouisson²

Table 2 Independent variables for hospital mortality

	<i>P</i>
SAPS 3 (per unit)	<0.001
ASA (per unit)	0.002
Fluid balance intraoperatively (per 100 ml)	0.006

BH: 1400 vs. 1950

Intraoperative fluids: how much is too much?



HIPO TENSÃO X Tx HEPÁTICO



HIPO TENSÃO IOP X TX HEPÁTICO

- IRA no POI Tx hepático – 40-80% dos pacientes!

Cirrose x
Vasoplegia

Hemorragia

Isquemia-
Reperusão

Imunossupressor

Manipulação
Cava

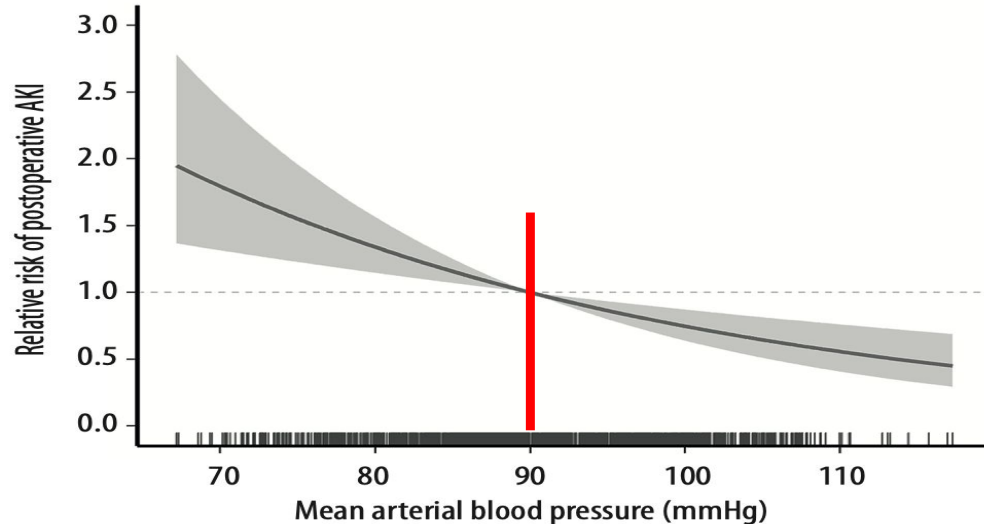
SCA



HIPO TENSÃO X IRA PÓS TX HEPÁTICO

**Low Mean Arterial Blood Pressure is
Independently Associated with Postoperative
Acute Kidney Injury After Living Donor Liver
Transplantation: A Propensity Score Weighing
Analysis**

A



PREVENIR ?



ELSEVIER

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American Journal of Transplantation

journal homepage: www.amjtransplant.org



Original Article

Higher vs standard mean arterial pressure target in the immediate postoperative period of liver transplantation to prevent acute kidney injury: A randomized clinical trial (LIVER-PAM)

Rodolpho Augusto de Moura Pedro ^{1,*} , Bruna Carla Scharanch ¹ ,
Lucas de Oliveira Araújo ¹ , Paula Sepulveda Mesquita ¹ ,
Luciana Severo Brandão ¹ , Lúcia da Conceição Andrade ² ,
Wellington Andraus ³ , Luís Augusto Carneiro D'Albuquerque ^{1,3} ,
Jean-Michel Constantin ⁴ , Alexandre Biasi Cavalcanti ^{5,6} ,
Alberto Queiroz Faria ⁷ , Luiz Marcelo Sá Malbouisson ^{1,3,5} 



PREVENIR ?

Randomização alvo PAM 24h pós op

PAM
65-70

PAM
85-90

Desfecho primário: Sem diferença na incidência de LRA em 7 dias.

MAKE28 (desfecho renal composto) foi menor no grupo intervenção



IRA perioperatória

- Comum e Mórbida !
- **Tratamento ?**
 - **Suspender nefrotóxicas**
 - **Evitar Hipotensão ?**
 - **Fluidos ?!**
 - **EGDT ?**
 - **TSR se necessário**
- **Prevenir ?**



Obrigada

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